

AUTACOIDS



Learning Outcomes

By the end of this lecture, students should be able to:

Recognize the pharmacologic effects of histamine

Describe the role of 5HT receptors in pain

Recognize the role of drugs acting on serotonin in prophylaxis and treatment of migraine and other types of pain

Describe the role of eicosanoids, kinins and related drugs

Introduction



Definition: Autocoids are biological factors (e.g., hormones, neurotransmitters) that act like local hormones and regulate physiological functions.



Types: Include histamines, serotonin, prostaglandins, bradykinin, and more.



Relevance to Dentistry: Role in inflammation, pain, and tissue repair, which are central to many dental procedures.

Classification of Autocoid Drugs

Major Classes:

- Histamine-related drugs
- Serotonin-related drugs
- Prostaglandins and leukotrienes
- Bradykinin
- Angiotensin

Functions in the Body: Autocoids help regulate blood flow, inflammatory response, pain, and smooth muscle contraction.

Histamine and Antihistamines

Role of Histamine:

- Vasodilation and increased vascular permeability
- Present in allergic reactions and inflammation

Receptors: H₁, H₂, H₃, H₄

H₁: Vasodilation, bronchoconstriction, itching

H₂: Gastric acid secretion

H₃, H₄: CNS and immune modulation

Antihistamines in Dentistry:

- H₁ blockers like diphenhydramine (for allergic reactions)
- H₂ blockers like Famotidine (for reducing gastric acid during surgery)

Prostaglandins (Eicosanoids) in Dentistry

Physiological Role:

- Inflammatory mediators
- Pain sensitization
- Regulate blood flow

Drugs Targeting Prostaglandins:

- **NSAIDs** (Non-Steroidal Anti-Inflammatory Drugs):
Inhibit cyclooxygenase (COX) enzymes to reduce inflammation and pain.
- **Examples:** Ibuprofen, Diclofenac

Clinical Relevance:

- Postoperative pain management
- Managing inflammatory conditions in oral tissues
- Misoprostol – gastric ulcer prevention, abortion

Leukotrienes



Role in the Body: Involved in bronchoconstriction, inflammation, and allergic responses.



Leukotriene Inhibitors:

Example: Montelukast
(used in asthma and allergic reactions)



Relevance in Dentistry: Limited use, but important for patients with asthma or allergic tendencies in response to anesthesia or other drugs.

Serotonin and Related Drugs

Role of Serotonin (5-HT):

- Vasoconstriction, platelet aggregation, regulation of mood and pain perception, peristalsis.

Drugs Targeting Serotonin:

- **SSRIs (Selective Serotonin Reuptake Inhibitors):** Primarily used in treating depression, but can impact dental treatments due to bleeding risks.

Dental Considerations:

- SSRIs can increase bleeding risk during surgery.
- Agonists: Sumatriptan (Migraine)
- Antagonists: Ondansetron (Antiemetic)

Other uses of 5HT (serotonin)

5 HT agonists

- **Sumatriptan:** Treat migraine attacks
- **Buspirone:** anxiolytics in anxiety disorders.

5 HT antagonists

- **Ondansetron:** Block vomiting centres and CTZ
- **Metoclopramide:** Antiemetic action and prokinetic

Bradykinin

Role in the Body:



Potent vasodilator and mediator of pain and inflammation. Increases vascular permeability and stimulates pain receptors.

Bradykinin Antagonists

Bradykinin acts on B₂ receptors, causing vasodilation, increased vascular permeability, and pain.

Icatibant competitively blocks B₂ receptors, thus preventing bradykinin's effects.

Rarely used directly in dentistry, but relevant in understanding pain and inflammation pathways.

Angiotensin and angiotensin converting enzyme (ACE) Inhibitors

Role of Angiotensin:

Regulates blood pressure by constricting blood vessels.

ACE Inhibitors:

Examples:
Lisinopril,
enalapril

May cause dry mouth (xerostomia) as a side effect.

Dental Implications: Dry mouth can increase the risk of dental caries and oral infections.

Clinical Applications of Autacoid Drugs in Dentistry

Pain Management: NSAIDs are frequently used to manage postoperative pain and inflammation.

Managing Allergic Reactions: Antihistamines play a critical role in treating allergic reactions to dental materials or medications.

Inflammation Control: Prostaglandin inhibitors (NSAIDs) are important for controlling inflammation and aiding in faster recovery.

Side Effects and Dental Considerations

NSAIDs: Gastrointestinal bleeding, renal impairment, and prolonged bleeding time.

Antihistamines: Xerostomia (dry mouth), drowsiness.

ACE Inhibitors: Dry mouth, which can lead to oral health complications.

Summary Table

Condition	Key Autacoid	Drug/Target
Allergy	Histamine	Antihistamines
Asthma	Leukotrienes	Montelukast
Peptic ulcer	Histamine (H ₂)	Ranitidine
Migraine	Serotonin	Sumatriptan
Hypertension	Bradykinin	ACE inhibitors (↑ bradykinin)

Understanding the role of autocoid drugs helps in the effective management of pain, inflammation, and other dental conditions

Essential Reading

- Bart –Jhonson, Frank J. Dowd. Pharmacology and Therapeutics for Dentistry, 8th edition. Elsevier Publishers, USA.
- Bertram G. Katzung et al. Basic & Clinical Pharmacology, 16th Edition, New York (NY): Lange / McGraw-Hill